

Course Title:	COE891
Course Number:	Software Testing and QA
Semester/Year (e.g.F2016)	W2024

Instructor:	Reza Samavi
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<i>Assignment/Lab Number:</i>	2
<i>Assignment/Lab Title:</i>	Coverage-based Test Design and Input Domain Model

<i>Submission Date:</i>	02/11/2024
<i>Due Date:</i>	02/12/2024

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*By signing above you attest that you have contributed to this written lab report and confirm that all work you have contributed to this lab report is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/current/pol60.pdf>

Lab Questions

Q1.

Elem		Cov%
▼ lab2		81.0%
▼ q1		99.6%
▶ IMoney.java		0.0%
▼ Money.java		97.9%
▶ Money		97.9%
▼ MoneyBag.java		100.0%
▶ MoneyBag		100.0%
▶ MoneyBagTest.java		100.0%
▶ MoneyTest.java		100.0%

Q2.

▼ q2		100.0%
▶ q2Function.java		100.0%
▶ q2tester.java		100.0%

Total branches:

1. $b > a$ is true (conditional branch)
2. $b > a$ is false (conditional branch). Then, $a > b$ is true (conditional branch)
3. $b > a$ is false (conditional branch). Then, $a > b$ is false (conditional branch)
4. Return statement at bottom (unconditional branch).

Total statements:

According to the lab manual, the total number of lines in the function is the total number of statements. Therefore, there are 10 total statements.

a = 2, b = 3:

$$\text{Statement Coverage} = \frac{\# \text{Executed Statements}}{\text{Total Number of Statements}} \times 100 = \frac{5}{10} \times 100 = 50\%$$

$$\text{Branch Coverage} = \frac{\# \text{Executed Branches}}{\text{Total Number of Branches}} \times 100 = \frac{1}{4} \times 100 = 25\%$$

a = 3, b = 2:

$$\text{Statement Coverage} = \frac{\# \text{Executed Statements}}{\text{Total Number of Statements}} \times 100 = \frac{5}{10} \times 100 = 50\%$$

$$\text{Branch Coverage} = \frac{\# \text{Executed Branches}}{\text{Total Number of Branches}} \times 100 = \frac{1}{4} \times 100 = 25\%$$

To cover all statements, this is the condition where $a = b$. For example, $a = 2$ and $b = 2$, however it can be any value as long as they are the same. This scenario was added and resulted in 100% coverage.

Q3.

Elem	Cov%
lab2	93.6%
q1	99.6%
q2	100.0%
q3	100.0%
Triclass.java	100.0%
TriclassTest.java	100.0%

Printed output:

```
Testing started
#Test_1 - started
#Test_1 - finished
#Test_2 - started
#Test_2 - finished
#Test_3 - started
#Test_3 - finished
#Test_4 - started
#Test_4 - started
#Test_5 - started
#Test_5 - finished
Testing is finished
```

